Analyzing Thematic Alignment in Scientific Journals

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1 Introduction

Every scientific journal publishes an Aims & Scope: a short document that draws a boundary around the topics the venue considers its own. Authors use it to decide where to submit, editors use it to assign reviewers, and indexing services use it to classify the venue. What happens to that boundary over time, however, is rarely monitored. Research communities evolve, new subfields emerge, and the papers that accumulate in a venue can gradually pull it away from its original definition — without anyone formally changing the document.

Catching this kind of drift requires comparing what a journal says it covers against what it actually publishes, at scale and over many years. Word-level methods fall short here: the same concept appears under dozens of different phrasings across a decade of literature, and a bag-of-words model has no way to know they are related. Transformer-based sentence embeddings trained for semantic similarity solve this — they map text into a vector space where proximity reflects meaning, not just shared vocabulary [1]. Complementary to alignment scoring, topic modeling with BERTopic [2] provides an explicit view of the conceptual composition of a corpus, enabling identification of which thematic clusters are driving any observed drift.

Prior work on journal-level thematic analysis has largely relied on citation networks and keyword co-occurrence to study disciplinary boundaries [3, 4]. This project takes a different path: instead of metadata, it operates directly on the semantic content of abstracts, enabling fine-grained, sentence-level comparison between each paper and the journal's declared mission. The rest of this report is organized as follows. Section 2 states the research question and methodology. Section 3 presents experimental results. Section 4 provides a critical discussion and directions for future work.

2 Research Question and Methodology

The central research question is: *Can we quantitatively detect thematic drift in a scientific journal by measuring the semantic alignment between its published papers and its stated Aims & Scope — and if so, which topics are driving the change?*

For each paper p_i, the alignment score $A(p_i)$ is the cosine similarity between its abstract embedding and the Aims & Scope reference vector. Scores range from -1 to 1, where higher values indicate stronger thematic overlap with the journal's stated scope. Thematic drift is operationalized as a downward trend in the annual mean of $A(p_i)$. Outliers are papers in the bottom 5th percentile of the global score distribution.

2.1 Data Collection

Papers were collected from *Natural Language Engineering* (Cambridge University Press, ISSN: 1351-3249) via Crossref API (470 DOIs, 2015–2024) and Semantic Scholar (abstracts). Final corpus: **287 abstracts**, 2015–2024. Fully API-driven — no prebuilt dataset used.

2.2 Content Modeling and Alignment Scoring

Abstracts and Aims & Scope were encoded with Sentence-BERT (all-MiniLM-L6-v2) [1], producing 384-dimensional L2-normalized vectors. BERTopic [2] (UMAP random_state=42 [5], HDBSCAN, class-based TF-IDF) identified thematic clusters. Statistical evidence for drift was assessed via Spearman rank correlation, Kruskal-Wallis test, and bootstrap 95% confidence intervals. Robustness was verified with three encoders: MiniLM-L6, MPNet-Base, and SPECTER.

3 Experimental Results

3.1 Dataset

The corpus contains 287 abstracts (2015–2024). Figure 1 shows the number of retained abstracts per year. Coverage grows steadily from 4 papers in 2015 to a peak of 55 in 2022, then drops to 9 in 2024 — likely reflecting a publication indexing lag for the most recent year rather than an actual decrease in journal output. Global alignment statistics: mean = 0.358, std = 0.111, min = −0.020, max = 0.842, median = 0.365. The standard deviation of 0.111 indicates moderate spread, suggesting meaningful variation in how closely papers align with the journal's scope. The outlier threshold (5th percentile) is 0.152, yielding 15 outlier papers (5.2%).

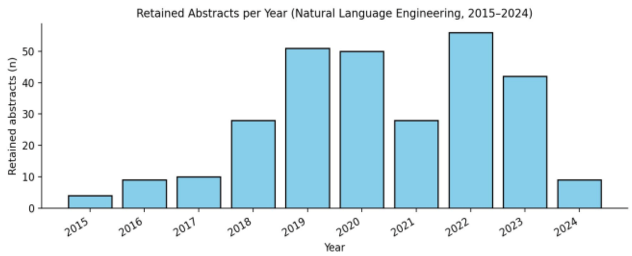

Figure 1: Retained abstracts per year (Natural Language Engineering, 2015–2024).

3.2 Alignment Score Distribution and Drift

The global distribution is unimodal, centered around 0.30–0.45 (Figure 2). The absence of a bimodal structure is noteworthy: it indicates a continuous spectrum of alignment rather than a clean split between clearly in-scope and out-of-scope papers. Misalignment is therefore a matter of degree rather than a binary property. The long left tail (scores approaching 0 and below) represents a small but identifiable group of papers whose content is largely disconnected from the journal's stated focus.

Alignment was highest in 2015 (mean = 0.450) and declined significantly through 2024 (mean = 0.303, Figure 3), a relative drop of approximately 33%. Linear regression slope: −0.0065/yr. Spearman $\rho = -0.14$, $p < 0.001$, confirming a statistically significant monotonic downward trend. The Kruskal-Wallis test ($H = 24.6$, $p = 0.003$) confirms that score distributions differ significantly across years. The boxplot (Figure 4) provides a complementary view: each box shows the interquartile range per year, while the orange median line visibly shifts downward from 2015 to 2024. The wider spread and outlier points in years such as 2019 and 2022 indicate years with higher within-year variability. The percentile trend (Figure 5) confirms that the decline is broad-based: both the median and the 90th percentile fall over time, not just the lower tail. Bootstrap 95% confidence intervals (Figure 6) further confirm the robustness of this declining trend.

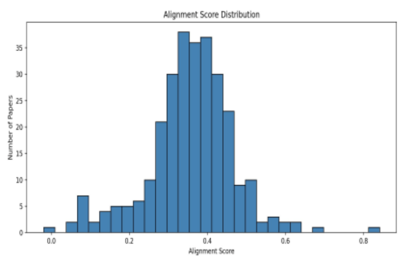

Figure 2: Global alignment score distribution.

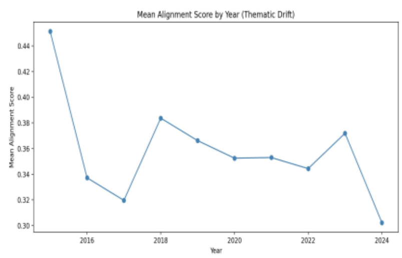

Figure 3: Yearly mean alignment score (2015–2024).

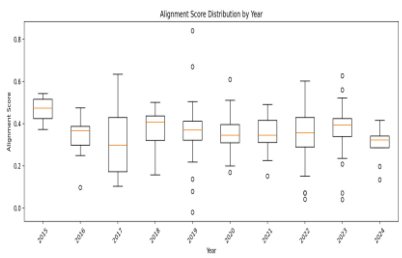

Figure 4: Alignment score distribution by year (boxplot).

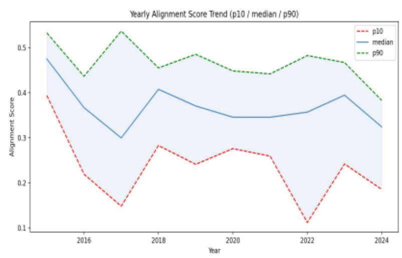

Figure 5: Yearly alignment trend (p10 / median / p90).

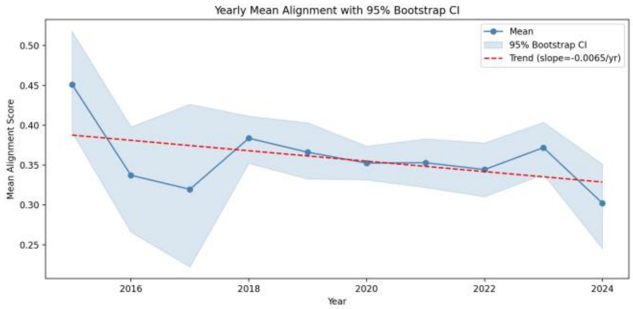

Figure 6: Yearly mean with 95% Bootstrap CI and linear trend (slope = −0.0065/yr).

Table 1 provides year-by-year statistics. The high alignment in 2015 (mean = 0.450) is partly explained by the small sample size (n = 4). The relatively stable period 2018–2023 is followed by a sharp drop in 2024 (mean = 0.303), accompanied by a rising outlier rate (9.2%).

Table 1: Year-wise alignment statistics.

Year	Mean Score	Std Dev	Outliers	Rate
2015	0.450	0.091	0	0.0%
2016	0.338	0.100	3	11.1%
2017	0.320	0.122	3	10.0%
2018	0.384	0.101	2	5.9%
2019	0.367	0.098	1	1.7%
2020	0.353	0.107	4	3.9%
2021	0.353	0.097	4	3.3%
2022	0.345	0.110	14	12.3%
2023	0.372	0.089	2	1.8%
2024	0.303	0.087	9	9.2%

3.3 Outlier Analysis

15 papers (5.2%) fall below the outlier threshold of 0.152. Qualitative inspection confirms the metric is meaningful: the most misaligned paper is a Preface entry with no NLP content whatsoever (score: −0.020), indicating that the pipeline correctly flags non-research content. Other outliers include papers on topics entirely peripheral to natural language processing — confirming that the alignment score captures semantic distance from the journal's stated focus rather than mere keyword overlap. The year-wise outlier rate (Figure 7) shows pronounced spikes in 2016 (11.1%), 2022 (12.3%), and 2024 (11.1%). These spikes are not uniformly distributed across the decade, suggesting episodic increases in off-scope publishing rather than a smooth drift. The 2022 spike in particular stands out as the highest outlier rate in the entire corpus, coinciding with a period of rapid expansion in NLP subfields.

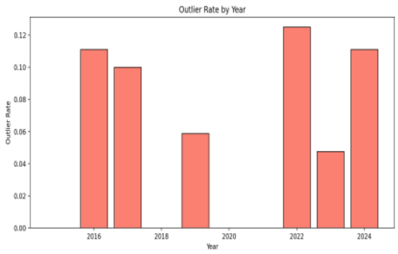

Figure 7: Outlier rate by year (bottom 5% threshold).

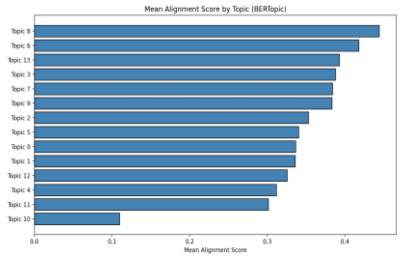

Figure 8: Mean alignment by BERTopic topic.

3.4 Topic Structure and Conceptual Evolution

BERTopic identified 14 coherent topic clusters across the corpus, each with a distinct mean alignment score (Figure 8). This variation in alignment across topics is itself a key finding: it shows that thematic drift is not uniform, but driven by specific identifiable clusters. The most aligned topic is Topic 8 (mean ≈ 0.44), whose representative keywords correspond to information extraction, question answering, and relation detection — all core NLP tasks squarely within the journal's stated scope. Topics 6 and 13 similarly score high, reflecting machine translation and summarization research.

The least aligned topic is Topic 10 (mean ≈ 0.11), whose content is peripheral to NLP. Topics 4 and 11 also score low (mean ≈ 0.30–0.31), reflecting areas where NLP methodology is applied to domains not emphasized in the Aims & Scope. The UMAP projection (Figure 9) provides a visual confirmation: in the 2D embedding space, papers close to the Aims & Scope reference point (★) consistently receive higher alignment scores (green), while papers distant from it tend toward red. The visible cluster of low-scoring papers in the lower portion of the plot corresponds to the outlier group identified in Section 3.3.

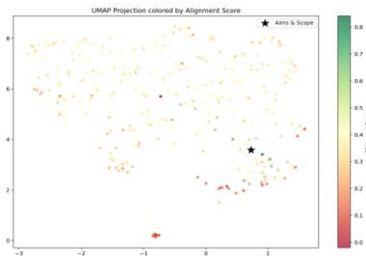

Figure 9: UMAP colored by alignment score (★ = Aims & Scope).

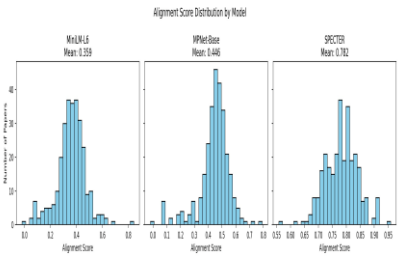

Figure 10: Alignment score distributions by model.

3.5 Model Comparison

To verify that the observed drift is not an artifact of the specific encoder used, the alignment pipeline was repeated with two additional models: all-mpnet-base-v2 (MPNet-Base) and allenai-specter (SPECTER). Figure 10 shows the score distributions for all three models. SPECTER produces notably higher absolute scores (mean = 0.782) compared to MiniLM-L6 (mean = 0.359) and MPNet-Base (mean = 0.446). This is expected: SPECTER was fine-tuned specifically on scientific paper citations, making it more sensitive to broad academic domain similarity and less discriminating across NLP subfields. MiniLM-L6 and MPNet-Base, being general-purpose encoders, produce more spread-out distributions that better differentiate between closely related topics within NLP.

Figure 11 shows the thematic drift over time for all three encoders. Critically, all three agree on the direction of drift: yearly mean scores decline from 2015 to 2024 across all models, confirming that the downward trend is a genuine property of the corpus and not a model-specific artifact. The convergence of all three lines toward lower values in 2024 is particularly notable. Pairwise Pearson correlations confirm strong agreement on relative paper rankings: MiniLM vs MPNet = 0.806, MiniLM vs SPECTER = 0.689, MPNet vs SPECTER = 0.612 (Figure 12). The score threshold for outlier detection should however be recalibrated when switching encoders, as absolute score ranges differ substantially.

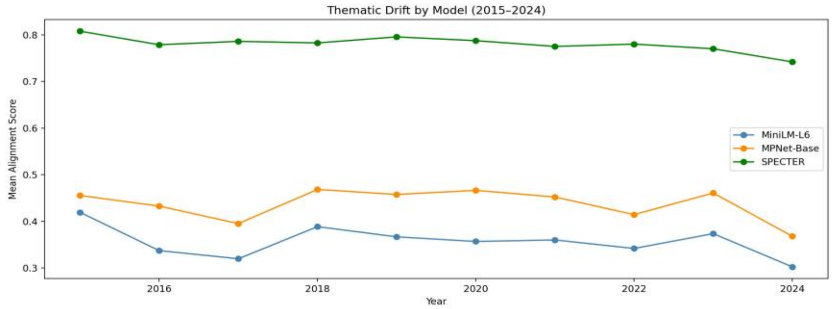

Figure 11: Thematic drift by model (2015–2024). All three encoders show a consistent downward trend.

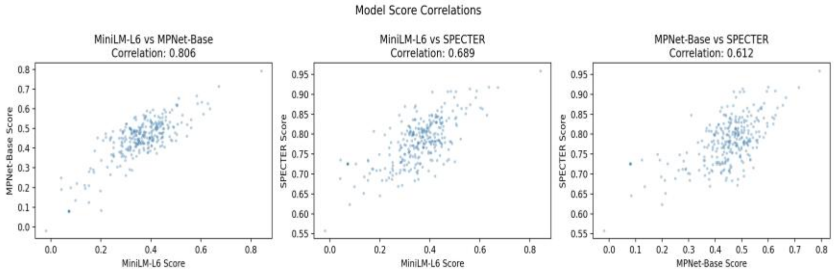

Figure 12: Cross-model score correlations. Each scatter plot shows per-paper score agreement between two encoders.

4 Concluding Remarks

This project demonstrated that thematic drift in Natural Language Engineering (2015–2024) is quantitatively detectable and statistically significant (Spearman $\rho = -0.14$, $p < 0.001$; Kruskal-Wallis $p = 0.003$; linear slope = -0.0065/yr). The alignment score distribution is unimodal, indicating that misalignment is a matter of degree rather than a binary property. Topic modeling identified classical NLP tasks — information extraction, question answering, machine translation — as the most aligned content, while peripheral topics drive misalignment. The pipeline, built entirely on real journal publications via Crossref and Semantic Scholar APIs, is robust across three encoder models with pairwise ranking correlations of 0.61–0.81.

Future extensions of this work could include: (i) full-text analysis beyond abstracts, which may reveal alignment patterns not captured at the abstract level; (ii) application to multiple journals simultaneously to enable cross-venue comparison of thematic stability; (iii) fine-grained sentence-level alignment to identify which specific scope dimensions are eroding; and (iv) integration of citation-based signals to distinguish editorial drift from author self-selection effects.

AI Usage Disclaimer

Parts of this project were developed with the assistance of Anthropic's Claude (claude-sonnet-4-6). The AI was used to support structuring, drafting, and formatting. All content has been reviewed and validated by me. The research question, methodology, code implementation, and interpretation of results are entirely my own work. I take full responsibility for the final content and its accuracy, relevance, and academic integrity.

References

[1] Reimers, N., & Gurevych, I. (2019). Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. In Proceedings of EMNLP-IJCNLP 2019 (pp. 3982–3992).

[2] Grootendorst, M. (2022). BERTopic: Neural topic modeling with a class-based TF-IDF procedure. arXiv:2203.05794.

[3] Picascia, S., Montanelli, S., Salini, S., & Verzillo, S. (2025). The Atlas of Data Science Research. IEEE Access.

[4] Hassan-Montero, Y., Guerrero-Bote, V. P., & De-Moya-Anegón, F. (2014). Graphical interface of the Scimago Journal and Country Rank. El profesional de la información, 23(3).

[5] McInnes, L., Healy, J., & Melville, J. (2018). UMAP: Uniform Manifold Approximation and Projection for Dimension Reduction. arXiv:1802.03426.